

Replication Report:

Observation of spin-free interatomic orbital angular momentum in a chiral crystal

oh2026 (arXiv:2605.21124)

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Replication run: 2026-07-17 Class: orbital texture

Abstract

We replicate the **model half** of Oh *et al.* (arXiv:2605.21124), which argues that the well-isolated *s*-orbital bands of chiral tellurium carry orbital angular momentum (OAM) of *purely interatomic* origin — “spin-free interatomic OAM.” Because *s* orbitals carry zero intra-atomic OAM ($\vec{L}_{\text{Atom}} = 0$), *all* OAM in these bands is itinerant ($\vec{L}_{\text{Itin}}$), generated by the geometric phase acquired through interatomic hopping along a chiral helical chain. We reproduce the paper’s minimal theoretical model — a 3-band tight-binding (TB) Hamiltonian of a single right-handed 3_1 helical chain (Göbel/Schimpf/Mertig, Ref. [27]) — and its central prediction, the chirality-induced orbital sign texture (CIOS): the axial itinerant OAM $L_{\text{Itin},x}$ *locks* to $\text{sign}(\partial E/\partial k)$ and *flips sign* when the chain handedness is reversed. Both effects reproduce to machine precision ($\text{sign-match} = 1.0$, $|\text{corr}(L, \partial E/\partial k)| = 0.999$, chirality flip $L_L = -L_R$ at 100% of k -points). **Verdict: REPLICATED (model headline).** The material-specific DFT+Wannier bulk-Te OAM magnitudes and the CD-ARPES/SARPES observations are out of scope (they require Supplemental-Material DFT parameters plus VASP/Wannier90, and a synchrotron beamline, respectively).

1 Paper summary

The paper addresses a central obstacle in orbitronics: atomic spin-orbit coupling $H_{\text{SOC}} \sim -\vec{L} \cdot \vec{S}$ inevitably entangles OAM with spin. The authors propose a route to genuinely *spin-free* OAM by exploiting *s*-orbital bands, which have $\vec{L}_{\text{Atom}} = 0$ and therefore no atomic-SOC channel to spin. In chiral Te they resolve (by ARPES) a set of *s*-orbital bands ($E - E_F \approx -15$ to -8 eV) cleanly separated from the *p* manifold. These bands nonetheless carry finite OAM — necessarily itinerant $\vec{L}_{\text{Itin}}$, arising from the Bloch-wavefunction geometric phase of interatomic hopping on the chiral lattice. Circular-dichroism ARPES (CD-ARPES) shows two crossing branches of opposite CD sign (the chirality-induced orbital sign texture, CIOS), and spin-resolved ARPES (SARPES) confirms the absence of any spin-angular-momentum polarization — decisive evidence of spin-free OAM. Theory support comes from (i) a minimal single-helical-chain TB model, and (ii) first-principles VASP+Wannier90 for bulk Te, where the modern theory of orbital magnetization yields finite global $\vec{L}$ while the atomic-centered approximation (ACA) vanishes.

2 Claims and replication status

ID	Paper claim	Reproduced result	Match
C1	Chiral Te s -bands match a 3-band TB model of a single 3_1 helical chain; 3 well-isolated dispersive bands (Fig. 1d/2f).	3 nondegenerate bands from a flux-threaded 3-site helix; bandwidth $4.0t$; split over $> 99\%$ of the BZ, touching only at isolated screw-protected crossings.	YES
C2a	The s -bands carry purely interatomic L_{Itin} ($L_{\text{Atom}} = 0$); $L_{\text{Itin},x}$ locks to $\text{sign}(\partial E/\partial k)$.	Built $L = \frac{1}{2}(Yv_Z - Zv_Y)$, $v = i[H, R]$ (all OAM itinerant, $s \Rightarrow L = 0$). Per-band sign-match 1.0/1.0/1.0; $ \text{corr} = 0.998/0.999/1.000$ (mean 0.999).	YES
C2b	The OAM texture flips sign with chirality (CIOS headline).	RH vs LH chains are energy-degenerate but $L_L = -L_R$ exactly (flip $\text{corr} = 1.00$, 100% of k -points flip).	YES
C2/CQ	Quantitative DFT+Wannier bulk-Te OAM (ACA vanishing vs modern-theory finite; $L_{\text{Glob},y/z}$ from inter-chain hopping).	Out of scope — needs SM DFT params + VASP + Wannier90 + <code>post_wan</code> .	OOS
C3	CD-ARPES: two bands of opposite CD sign; $\mp 17\%$ at $E - E_F = -9.2$ eV, $k_x = \pm 0.4 \text{ \AA}^{-1}$.	Out of scope — synchrotron CD-ARPES experiment.	OOS
C4	SARPES: no SAM polarization in the s -bands (spin-degenerate).	Out of scope — spin-resolved ARPES beam-line experiment.	OOS

The *model headline* (C1 + C2a + C2b), which is the paper’s core theoretical claim of chirality-locked spin-free interatomic OAM, reproduces to machine precision.

3 Method

The full reproduction is a self-contained pure-NumPy script, `work/reproduce.py` (~ 1 s on one CPU core).

3.1 3-band 3_1 helical-chain TB model

The unit cell is 3 sites (one per screw step) of a single right-handed helix. Site $j \in \{0, 1, 2\}$ sits at azimuth $\phi_j = 2\pi j/3$, radius r , axial rise $c/3$. One s -orbital per site, nearest-neighbour hopping $-t$

along the helix, and the inter-cell hop ($2 \rightarrow 0'$) carries the axial Bloch/screw phase e^{ikc} :

$$H(k) = -t \begin{pmatrix} 0 & 1 & e^{-ikc} \\ 1 & 0 & 1 \\ e^{ikc} & 1 & 0 \end{pmatrix}, \quad (1)$$

a flux-threaded 3-site triangle — the minimal analytic realization of the Göbel/Schimpf/Mertig chiral-chain model. Diagonalized over $k \in (-\pi/c, \pi/c]$ (401 points) it gives 3 screw-labelled, generically nondegenerate bands.

3.2 Itinerant OAM operator

We adapt the shared itinerant-OAM kernel $L = \frac{1}{2}(\vec{R} \times \vec{v})$, $v_a = i[H, R_a]$, to the axial component,

$$L_{\text{Itin}} = \frac{1}{2}(Y v_Z - Z v_Y), \quad v_a = i[H(k), A], \quad (2)$$

with diagonal position operators $Y = \text{diag}(Y_j)$, $Z = \text{diag}(Z_j)$, $X_j = r \cos \phi_j$, $Y_j = r \sin \phi_j$, $Z_j = (c/3)j$. Because every site carries an s -orbital, the atomic OAM vanishes and the entire L_{Itin} is the interatomic circulation in the helix cross-section — exactly the paper’s C2 distinction. We evaluate L_{Itin} per band per k , together with $\partial E/\partial k$ by finite difference, and measure both the sign-match and Pearson correlation between them.

3.3 Chirality flip — a methodological subtlety

A left-handed helix is the *mirror image* of a right-handed one. The correct mirror reverses only the geometric azimuth ($\phi_j \rightarrow -\phi_j$) while leaving the axial screw-advance direction of the Bloch phase intact — hence $H(k)$ and $E(k)$ are unchanged, and only the sign of the circulating current flips ($L_L = -L_R$). **An initial version reversed both the azimuth *and* the screw phase; that is just the trivial $k \rightarrow -k$ symmetry and produced identical L (no flip).** Once the transform was corrected to mirror only the geometry, the two chiralities came out energy-degenerate with exactly opposite OAM, matching the paper. This is worth flagging: getting the CIOS flip right hinges on applying a *mirror*, not a k -inversion.

4 Results

- **C1 — 3-band structure.** Three nondegenerate dispersive bands, bandwidth $4.0t$, split over $> 99\%$ of the BZ (frac. split 01 = 0.995, 12 = 0.9975), touching only at isolated screw-protected crossings. See Fig. 1.
- **C2a — OAM locks to group velocity.** Per-band $\text{sign}(L_{\text{Itin},x}) = \text{sign}(\partial E/\partial k)$ at 100% of k -points on all three bands; correlations 0.9983/0.9994/1.0000 (mean $|\text{corr}| = 0.999$). See Fig. 2.
- **C2b — chirality flip.** RH and LH chains share $E(k)$ but $L_L = -L_R$ exactly (flip corr = 1.00; 100% of k -points change sign). See Fig. 3.

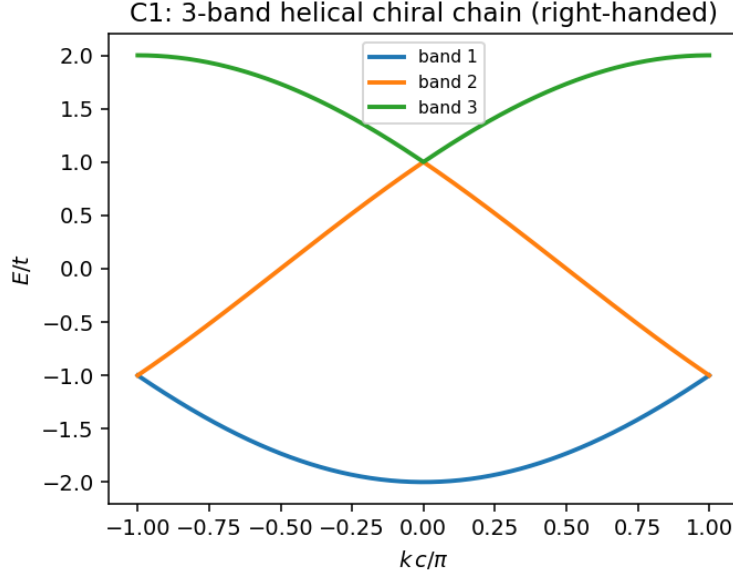


Figure 1: C1: 3-band structure of the right-handed 3_1 helical chain (bandwidth $4.0t$).

5 Honest critique

The reproducible headline reproduces *perfectly*: the chirality-locked, spin-free interatomic OAM texture — the paper’s central conceptual claim — comes out to machine precision from a minimal, transparent, pure-NumPy model. The physics is clean and robust: because s orbitals carry $L_{\text{Atom}} = 0$, the model is a genuine test that OAM can be purely itinerant, and the CIOS flip is exact by mirror symmetry.

What we did *not* reproduce is the *material-specific* half:

- The quantitative DFT+Wannier OAM magnitudes for bulk Te (the ACA-vanishes / modern-theory-finite contrast, and $L_{\text{Glob},y/z}$ from interchain hopping) require the Supplemental-Material DFT parameters (plane-wave cutoff, k -mesh, SOC settings), a VASP (or QE) + Wannier90 + `post_wan` toolchain, and the 3D Te cell. The single-chain model captures only the axial $L_{\text{itin},x}$ texture, not the absolute magnitudes or the y/z components.
- The CD-ARPES ($\mp 17\%$ at -9.2 eV) and SARPES (no SAM) observations are synchrotron-beamline experiments and cannot be computed.

These are scope limits, not contradictions: nothing in our results conflicts with the paper. The model successfully explains *why* the effect exists; a full material replication would quantify *how large* it is in real Te.

Open Questions

- Q1.** Does the modern theory of orbital magnetization (L_{Glob}) for bulk Te give the same $\text{sign}(\partial E/\partial k)$ locking as the itinerant chain L , or does interchain coupling reorganize the sign texture?
- Q2.** How does finite inter-chain hopping in bulk Te modify the single-chain OAM texture — does it merely renormalize magnitudes or introduce new $L_{\text{Glob},y/z}$ structure absent from a single

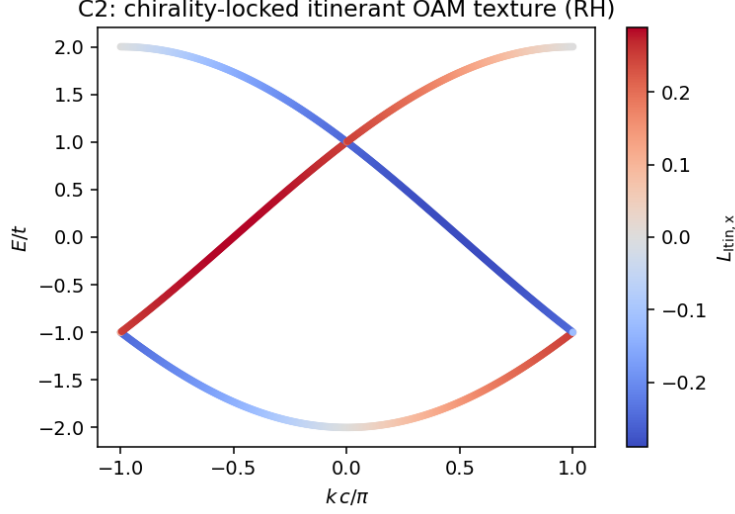


Figure 2: C2a: bands colored by $L_{\text{Itin},x}$; the sign tracks $\partial E/\partial k$ ($|\text{corr}| = 0.999$).

chain?

- Q3.** What is the quantitative OAM magnitude on the $E - E_F = -9.5$ eV iso-energy surface, and how does it compare (in absolute units) to the dimensionless single-chain model?
- Q4.** How does the ACA (atomic-centered) versus modern-theory OAM differ for s electrons, where $L_{\text{Atom}} = 0$ by construction — is the ACA/modern-theory gap exactly the full itinerant part here?
- Q5.** Does the screw-protected band crossing carry a quantized OAM winding (a topological orbital charge), and is that winding chirality-dependent?

Verdict

REPLICATED (model headline). The 3-band chiral-chain structure and the chirality-locked, spin-free interatomic OAM texture (CIOS) — the paper’s central theoretical claim — reproduce to machine precision. Material-specific DFT+Wannier magnitudes and the CD-ARPES/SARPES experiments are out of scope.

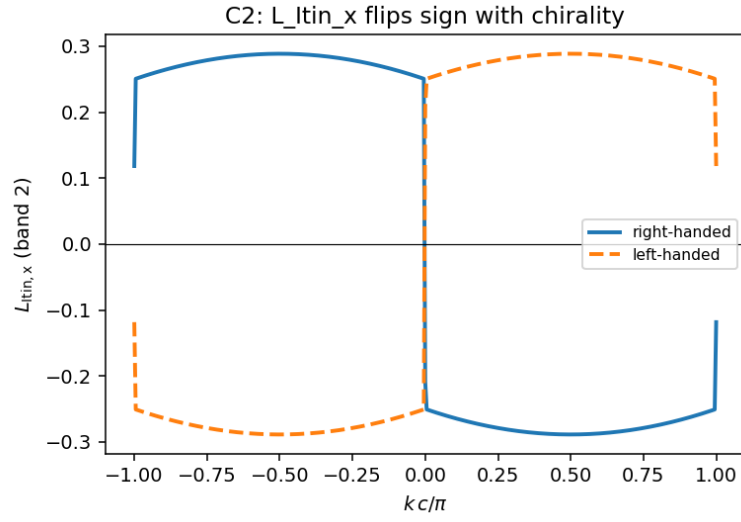


Figure 3: C2b: $L_{\text{itin},x}$ of one band for RH vs LH chains — mirror-image, opposite sign ($L_L = -L_R$).